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Non-contact breathing monitoring by integrating RGB and thermal imaging via RGB-thermal image registration

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ABSTRACT

Altered breathing rate is an important sign of disease status. Currently used machine-based monitoring of the breathing rate includes contact with the body, which may result in pain and discomfort. In this paper, a non-contact breathing monitoring technique is proposed by integrating RGB and thermal imaging systems with RGB-thermal image registration. This method provides a linear mapping for automated selection of the region of interest (ROI) followed by tracking to extract the breathing rate. To evaluate the efficacy of the proposed approach and its robustness against motion, talking, varying breathing rate or rhythm, and high ambient temperature, this study was conducted in three phases. Validation of the proposed approach demonstrated a strong agreement with the reference method of breathing rate monitoring using a respiration belt. During normal breathing, the mean absolute error (MAE) reached 0.11 bpm (breaths per minute). While in more challenging conditions, defined by three phases, the MAE reached 1.46, 2.08, and 1.69 bpm, respectively. In short, the proposed method performance is a promising alternative to a contact-based method due to its strong agreement and might be useful in diverse applications such as sport studies, rehabilitation centres, quarantine centres, and in hospital or airport screening during the COVID 19 pandemic.

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1. Introduction

Breathing rate (or respiration rate), along with body temperature and heart rate, is one of the three vital indicators of the health condition of humans. Breathing consists of inspiration, which involves the inhalation of oxygenated air, and expiration, which involves the expulsion of carbon dioxide-rich air. The phase of inspiration and expiration makes a cycle, and the number of cycles in a minute is referred to as the breathing rate (BR). For adults under resting conditions, the BR varies from 12 to 20 bpm (breaths per minute) [1]. Normal BR is faster in children with a newborn infant normally breathing at 30–60 bpm [2]. BR is a sensitive predictor of abnormal health condition and need of admission to the hospital [3,4]. A variety of diseases cause abnormal breathing dynamics, which can be detected by changes in breathing rate, breathing sound, or breathing pattern. Moreover, the BR is also important in many other areas and research, including emotion classification [5], robotic rehabilitation [6], influenza detection at airports [7], respiratory infections [8], and sport studies [9]. BR has been termed as 'the neglected vital sign' [4] despite its importance in medical diagnosis and scope in various research areas [10]. Presently, almost all BR monitoring methods require contact with the body, which induces pain or discomfort. Moreover, the dislocation of adhesive sensors compromises the measurement quality or, in the worst case, damages the skin in patients with sensitive skin such as infants [11] or burn patients. Additionally, the replacement of disposable sensors (e.g., an electrode), handling, and hygiene are more complex and expensive, and pile up medical waste.

Recently, researchers have shown increased interest in unobtrusive and non-contact based vital sign detection methods. Numerous non-contact techniques have been proposed, including those based on chest wall movement detection, such as radar based [12], depth sensing [13,14], and those based on imaging techniques, such as visible camera [15], remote-photoplethysmography (r-PPG) [16], and thermal imaging [17]. Besides offering contactless measurement, the radar-based method is still limited because of the risk of radiation exposure and is sensitive to motion artefacts. On the other hand, the camera based and remote PPG based methods are sensitive to the illumination. Nonetheless, this area is undergoing continuous improvement [18,19].

Compared to the above methods, thermal imaging can be considered more suitable due to its non-contact, non-invasive, and non-radiative nature. Thermal imaging, alternatively referred to as infrared thermography (IRT), is the visual depiction of an object's infrared energy emission. A thermal camera detects the infrared energy and produces a black and white or pseudocolor thermal image. Breathing measurement using thermal imaging is based on the detection of temperature changes of nostrils and/or mouth during inspiration and expiration. This is because the temperature of the inhaled air is at the same to the ambient temperature, whereas the exhaled air temperature is nearer to the body temperature. In comparison to the chest wall movement detection used in radar-based and camera-based methods, the temperature variation in the nostril area is a more reliable strategy for breathing monitoring.

Zhu et al. [20] developed a novel non-contact method for breathing monitoring by manually selecting three facial windows, two at the periorbital region, and one at the nasal apex. The breathing signal was extracted by tracking of the rectangular window placed on the nose. In [21], the temperature variation recorded by the thermal camera from the side view of the subject was used to detect the breathing rate. Abbas et al. [22] proposed a method of breathing rate measurement using the manually selected region of interest (ROI) in thermal images and also tested on the infant. AL-Khalidi et al. [23] used a Sobel edge detection method to identify the boundary of the subject's head in the threshold image followed by superimposing the ellipse on the segmented image and scanning for the warmest and coolest regions. The warmest regions corresponded to the bridge of the nose and the medial aspect of the eye, and the coolest region corresponded to the tip of the nose. Pereira et al. [17] used the multilevel Otsu's method to extract the largest region in an image and considered it to be the subject's face. After that, the periorbital region, the hottest area of the face because of its vascular anatomy, was selected to find the ROI. Further, Pereira et al. [24] extended the work by considering the temperature variation not only in the nostril but also in the mouth area. The movement of both shoulders was also incorporated for BR extraction. The data of these regions are fused to extract the BR, which further improves the accuracy. However, the salient physiological features of the human face in thermal images used to find the ROI have not been tested in more practical scenarios, such as wearing glasses, with facial hair (beard and moustache), talking, and high thermal dynamic range scenes. In these cases, the image segmentation might not effectively segment out the face and nostril area, leading to erroneous ROI selection. Cho et al. [25] introduced the 'optimal quantization' and the 'thermal gradient flow' to improve the accuracy of nostril tracking. Also, a thermal voxel method was introduced in place of the averaging method. The 'Ensemble of Regression Trees' was used by Jagadev et al. [26] for automatic tracking of manually selected nostrils in thermal images.

In comparison to thermal images, the visible images provide richer data of the subject's face due to the red (R), green (G), and blue (B) spectrum illumination. Accordingly, it is also referred to as an RGB image. It has more detail and texture information, and has a higher spatial resolution than the corresponding thermal image, which has only a single intrinsic property of the face, namely radiated heat [27]. Also, the thermal image is limited in spatial resolution, while high-resolution thermal cameras are costly. The visible images are more accurately used for feature selection, classification, and tracking in comparison to thermal imaging. Wang et al. [28] proposed a novel approach for extracting the salient facial features such as the corners of the eyes and mouth, as well as the nostrils, from thermal images by fusing visible and thermal imaging systems. This method provides an automatic non-contact method to precisely locate the facial features in thermal images to read the face temperature of the

ROI area. Scebba et al. [29] introduced the multispectral camera fusion to improve the ROI detection in thermal imaging. Further, Scebba et al. [30] used the multispectral ROI detection strategy for respiratory rate and apnea monitoring by combining near-infrared (NIR) and far-infrared (FIR) cameras. Hu et al. [31] presented a synergetic use of thermal and RGB imaging to quantify breathing in a non-contact manner. However, these approaches specified a fixed dual imaging configuration, in which the two cameras were permanently attached to one another. From the first frame of both thermal and visible video, the match control points were selected to obtain the registration parameters, which were applied to analyse all the recording datasets. These parameters are much dependent on the subject-to-camera distance and the camera setup [29,31].

In the multispectral camera-based method, the first step is the alignment of images in the same coordinate so that complementary features can be used to improve the general robustness of camera-based BR monitoring. In general, the registration is performed manually by the first frame of recorded videos, which limits the performance due to the fixed camera settings. This paper introduces an automatic registration process in a multispectral camera-based method to avoid the limitation of fixed camera settings. Automatic RGB-thermal image registration algorithms provide a more realistic, effective, and accurate process, and many methods have been proposed recently. However, due to the multimodality and texture differences in RGB-thermal image pairs, the automatic registration of RGB and thermal images presents several challenges. The most challenging steps in the automated multispectral registration process are feature extraction and matching. To overcome these challenges, frequency based (structural element based) feature extraction and description methods have been proposed by researchers. The paper [32] presented an edge oriented histogram for feature matching in thermal and visible image registration which is described by the shape and contour of the neighbourhoods. Mouats et al. [33], proposed a phase congruency based feature extraction and edge histogram descriptor (PCEHD) for thermal-visible image registration. Aguilera et al. [34] proposed a Log-Gabor filter based feature descriptor with multiple scales and orientations. Ye et al. [35] employed the magnitude and orientation of phase congruency to determine the well-normalized local histogram for feature descriptors. Li et al. [36] proposed an insensitive to radiation change based image registration method to overcome the gradient constraint on feature description in multi-mode images.

Inspired by [31,37], we proposed a modular framework for the non-contact breathing rate detection method by integrating RGB and thermal imaging systems. This includes: an RGB-thermal image registration to obtain the registration parameters for linear mapping of RGB to thermal images; the face and facial landmark detection and tracking to obtain the temperature variation; and validation of the proposed modular system for breathing rate and pattern monitoring. The following section 2 explains the steps involved in the proposed method. Section 3 provides the experiment setting, protocols, and evaluation metrics for validation. Section 4 and 5 describe the results and discussion, respectively. Finally, section 6 explains the conclusion and future direction.

2. Proposed methodology

The basic principle of thermal imaging for detecting breathing rate is that the temperature varies around the nostrils and mouth area during the breathing process. In general, the nostril area (ROI) is selected and tracked over successive video frames, and the average of pixels value within the ROI gives the breathing signal. However, as explained earlier, thermal images have less texture and depth of information than visible images, making facial feature detection and tracking algorithms in thermal imaging ineffective. To address this issue, the RGB image sequence is integrated with the thermal image sequence for the purpose of precisely detecting and tracking facial features. The overall block diagram of the proposed modular system is shown in Fig. 1. The consequential steps of development are explained in the following subsections.

2.1. *RGB-Thermal acquisition*

The first step is the simultaneous acquisition of thermal and RGB image sequences. The acquisition setup has two cameras - a thermal camera and a visible camera. These are placed in parallel in such a way that their field of view (FOV) covers the same scene, and the distance between them is less so that they adhere to the principle of the pinhole camera model. The setting parameters and camera specifications are explained in the Experimental setup and protocols subsection.

2.2. *RGB-thermal image registration*

Due to the difference in camera parameters like focal length, pixel size of the sensor, etc., the acquired visible and thermal image sequences exhibit large scale differences. In the proposed methodology, first, a scale adjustment pre-transformation is applied to align the visible image into the same coordinates of the thermal image. Then a fine registration algorithm is performed for precise alignment, as shown in Fig. 1.

In pre-transformation, a calibration rig is used to find the common correspondence points in visible-thermal image pairs. A calibration rig is an easy-to-make type of structure and is made by cutting a regular-sized square from a thin material that is a poor conductor of heat [38]. The subject (human) used this rig at the initial time of video acquisition (as shown in Fig. 1). The starting frames of the videos with calibration rig are used for the pre-transformation algorithm. The algorithmic steps involved in the pre-transformation process are shown in Fig. 2. The basic ideology of using a calibration rig is to extract the control points automatically for the transformation matrix. Although the pre-transformation scales the RGB image into the same coordinate as the thermal image, there might be some misalignment due to the limitation of control points selected in a specific (calibration rig) area and uniformly not distributed in the image. A fine registration algorithm inspired by the phase congruency method for feature selection is also applied to refine the alignment. In the phase congruency method, the image features are

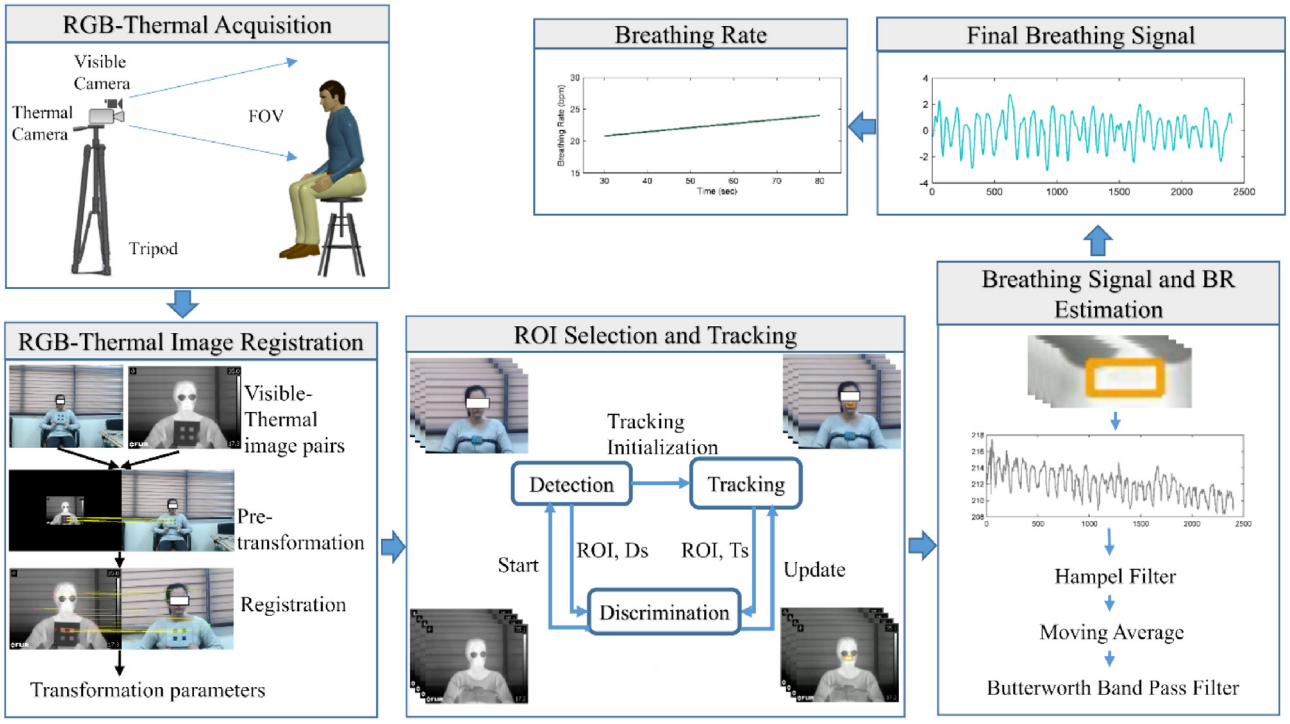


Fig. 1 – The overview of the proposed method.

defined as the points in the image where all the Fourier components are in phase [39]. As a result, this approach operates in the frequency domain and uses structural elements such as corner and edge points, which is more efficient and robust compared to intensity or gradient-based feature extraction methods. Kovesi [40], described the use of quadrature pair of log-Gabor wavelets for phase congruency calculation. The steps involved in the fine registration process are feature extraction, feature descriptor, feature mapping, and transformation.

Feature Extraction: By considering F_{so}^{even} and F_{so}^{odd} as the even and odd components of log-Gabor filter at scale s and orientation o, the even and odd responses of input image $I(i,j)$ can be expressed as:

$$[E_{so}(i,j), O_{so}(i,j)] = [I(i,j) * F_{so}^{even}, I(i,j) * F_{so}^{odd}] \tag{1}$$

From the even and odd responses $(E_{so}(i,j), O_{so}(i,j))$, the amplitude component $A_{so}(i,j)$ and phase component $\phi_{so}(i,j)$ of an input image $I(i,j)$ can be calculated as:

$$A_{so}(i,j) = \sqrt{E_{so}(i,j)^2 + O_{so}(i,j)^2} \tag{2}$$

$$\phi_{so}(i,j) = arctan\left(\frac{O_{so}(i,j)}{E_{so}(i,j)}\right) \tag{3}$$

According to the Kovesi [40], the phase congruency for each orientation o can be calculated by using following equation:

$$PC_o(i,j) = \frac{\sum_s w_o(i,j)\lfloor A_{so}(i,j)\Delta\Phi_{so}(i,j) - N\rfloor}{\sum_s A_{so}(i,j) + \varepsilon} \tag{4}$$

where $o = 1, 2, \cdots N_o$ and $s = 1, 2, \cdots N_s$ represent the number of orientations and scales. The $w_o(i,j)$ is a factor that measures the weight of the frequency spread. A small positive real value ε is used to avoid division by zero and N is the estimated noise influence. The operator $\lfloor\rfloor$ represents the enclosed quantity is either the value itself or zero if it is less than zero. $A_{so}(i,j)\Delta\Phi_{so}(i,j)$ is calculated as:

$$A_{so}(i,j)\Delta\Phi_{so}(i,j) = E_{so}(i,j)\overline{\phi}_o^{even}(i,j) + O_{so}(i,j)\overline{\phi}_o^{odd}(i,j) - \left|E_{so}(i,j)\overline{\phi}_o^{odd}(i,j) - O_{so}(i,j)\overline{\phi}_o^{even}(i,j)\right| \tag{5}$$

where,

$$\overline{\phi}_o^{even}(i,j) = \sum_s E_{so}(i,j)/C_o(i,j) \tag{6}$$

$$\overline{\phi}_o^{odd}(i,j) = \sum_s O_{so}(i,j)/C_o(i,j) \tag{7}$$

$$C_o(i,j) = \sqrt{\sum_s E_{so}(i,j) + \sum_s O_{so}(i,j)} \tag{8}$$

The above steps are used to find the phase congruency $PC(\theta)$ over multiple scales in a given orientation θ. By following the classic moment analysis equations [41], the maximum moment M and minimum moment map m are calculated as follows:

$$M = \frac{1}{2}(c + a + \sqrt{b^2 + (a-c)^2}) \tag{9}$$

$$m = \frac{1}{2}(c + a - \sqrt{b^2 + (a-c)^2}) \tag{10}$$

Algorithm: Pre-Transformation

Input: A thermal-visible image pair (I_{ir} , I_{vis})

Output: Pre-transformation matrix $\boldsymbol{T}$, coarsely align visible image I_{vis_s}

1. Generate a binary image pair (BI_{ir} , BI_{vis}) by applying Otsu thresholding on (I_{ir} , I_{vis})

$$BI_{ir} = \begin{cases} 1 & if\ I_{ir} \le T_{ir} \\ 0 & otherwise \end{cases} \quad ;$$

$$BI_{vis} = \begin{cases} 1 & if\ I_{vis} \le T_{vis} \\ 0 & otherwise \end{cases}$$

where, T_{ir} and T_{vis} are the threshold value of I_{ir} and I_{vis} respectively.

2. Detect every connected region in each binary image (BI_{ir} , BI_{vis}) and calculate the feature of detected region like major_axis, minor_axis, area and perimeter.
3. For each detected region in (BI_{ir} , BI_{vis}) calculate two selection criteria as follow:

$$c_1 = \frac{\sqrt{area}}{0.25 \cdot perimeter} \quad ; \quad c_2 = |major_{axis} - minor_{axis}|$$

4. To select the two approx square size region in both BI_{ir} and BI_{vis} , apply the two filtering criteria as follow:

$$0.9 < c_1 < 1.1 \quad ; \quad c_2 < 3$$

5. After filtering, the squares formed due to calibration rig in both BI_{ir} and BI_{vis} are separated out, and their centroid and top left corner are used as correspondence pairs in input image pair (I_{ir} , I_{vis}).
6. By assuming the fixed-point x in I_{ir} and their correspondence point x' in I_{vis}, the affine transformation matrix can be calculated by solving following relationship.

$$\boldsymbol{x'} = \boldsymbol{T}\,\boldsymbol{x}$$

7. Align the I_{vis} into the same coordinate of I_{ir} according to

$$I_{vis_s} = I_{vis}\boldsymbol{T}$$

Fig. 2 – The algorithmic steps in pre-transformation.

where:

$$a = \sum (\mathrm{PC}(\theta)\cos(\theta))^2 \tag{11}$$

$$b = 2\sum (\mathrm{PC}(\theta)\cos(\theta)) \cdot (\mathrm{PC}(\theta)\sin(\theta)) \tag{12}$$

$$c = \sum (\mathrm{PC}(\theta)\sin(\theta))^2 \tag{13}$$

The corner feature points are extracted using filtering based on non-maxima suppression and thresholding on the minimum moment map. The edge feature points are extracted by applying a FAST [42] feature detection method on the maximum index map. Calculated significant corner and edge feature points are used for feature matching.

Feature Descriptor: A feature descriptor is required to distinctly identify the feature point. Inspired by [36], the log-Gabor convolution sequences are used to construct the feature descriptor. The amplitudes $A_{so}(i,j)$ of N_s scales are added for each orientation o,

$$A_o(i,j) = \sum_{s=1}^{N_s} A_{so}(i,j) \tag{14}$$

A cell of N_o layer is represented as $\{A_o(i,j)_1^{N_o}\}$. The maximum value of array $A_o(i,j)_1^{N_o}$ for each pixel location (i,j) can be called as maximum index map (MIM). After finding MIM, at each extracted features points, a local patch of $P \times P$ is selected. Further, it is divided into sub-blocks of 6×6 and constructed a histogram with bin N_o for each sub-block. Accumulate all the histograms to construct a feature descriptor for each extracted feature point which has a dimension of $6 \times 6 \times N_o$.

Feature Matching and Transformation: The sum of square differences (SSD) is used to find the matched feature points. The two feature points defined by the feature descriptor fd_1 and fd_2 in image $I1$ and $I2$ respectively, are matched when they satisfy the following criteria.

$$SSD(fd_1, fd_2) < d_{ratio} \cdot SSD(fd_1, fd'_2) \tag{15}$$

where fd'_2 is nearest neighbour to fd_1 in the image $I2$. Finally, the fast sample consensus (FSC) [43] is applied to find the transformation parameters from the matched corresponding keypoints.

2.3. ROI selection and tracking

Face and facial landmark detection and tracking are standalone problems in computer vision and are employed in various applications like security, recognition, deception, medical, etc. As discussed earlier, the facial landmark detection for ROI selection in RGB images is more robust than thermal imaging. The ideology of this work is that the ROI selection and tracking in the visible spectrum may be easily mapped into the thermal domain using the obtained transformation parameters. The benefit of this approach is the elimination of the need for detection and tracking in thermal images, which have less texture than visible images.

In this work, we implemented face landmark detection using the multi-task cascaded convolution neural network (MTCNN) designed and trained by Zhang et al. [44]. MTCNN is faster and has better performance in comparison to traditional methods. It can robustly identify five landmarks- centre of left and right eye, peak of nose, right and left mouth corner. The ROI is defined by the rectangular bounding box. ROI containing the horizontal and vertical coordinate (X, Y), width W and height H, as described below:

$$ROI = [X, Y, W, H] = [nx - w, ny - h, 2 \cdot w, 3 \cdot h] \tag{16}$$

where

$$h = \frac{\sqrt{(mlx - nx)^2 + (mly - ny)^2}}{4} \tag{17}$$

$$w = round\left(\frac{|mlx - mrx|}{2}\right)$$

The (nx, ny), (mlx, mly) and (mrx, mry) are the coordinates of the peak of the nose, left mouth corner and right mouth corner, respectively.

The tracking of ROI is the localization of the ROI over the sequence of frames. The correlation filter (CF) trackers are accurate, fast and mostly used for object detection [45]. CF trackers learn the correlation filter according to the information of the initial and next frame that can be applied to the next frame for location update (location of maximum filter response). However, there are many drawbacks associated with CF trackers due to the drift in fast motion, background clutter, occlusion, and boundary effects. In the proposed method, the nose region has to be tracked over the frame sequences. Since the desired object is small in comparison (has less pixels), tracking can fail or become less precise. Also, in the case of long-time tracking, drift may happen. Therefore, a tracking-by-detection (TD) approach is applied in this work to track the nose region. In the TD framework, the combination of tracking and detection, which is connected by the discrimination, is used to update the ROI location. Here, the discriminative scale space tracker (DSST) [46] is connected with the multi-task cascade CNN detector [44]. The primary function of discrimination is to enable the tracking model to be learned by introducing detection when tracking fails. The pseudocode of the tracking-by-detection approach is shown in Fig. 3. The final result $R(k)$ represents the ROI image for each frame k. This approach eliminates any drift or tracking failures, enabling for more accurate recording of the temperature profile enclosed inside the tracked ROI in successive frames.

2.4. Breathing signal and BR estimation

In each frame k of the video sequence, the average of pixel intensity within the track ROI $R(k)$ is calculated according to:

$$\bar{s}(k) = \frac{1}{M \times N} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} TI_{ROI}(x, y, k) \tag{18}$$

where $TI_{ROI}(x, y, k)$ denotes the intensity values of thermal image TI within ROI at frame k. M and N are the width and height of ROI, respectively. In brief, the raw breathing signal $\bar{s}(k)$ is obtained by the average temperature of ROI for all frames. The Hampel filter is used to eliminate some unwanted spikes in the obtained respiration signal. In this filter, successive windows of the obtained respiration signal are compared with the median absolute difference to remove the outlier [47]. Further, the extracted breathing signal is smoothed by the moving average filter to mitigate the abrupt changes. After that, the desired breathing signal is extracted by applying a Butterworth band pass filter with the 3 dB cutoff frequency in the range of [0.1, 0.85]. The proposed method uses a chirp z-transform (CZT) [48] for respiratory rate calculation. The advantage of CZT is that it can focus on a specific frequency range [49]. The frequency resolution can be determined from the limited frequency bandwidth of the breathing signal.

3. Experimental setup and protocol

The experimental setup of the proposed method is shown in Fig. 1. The camera setup is placed approx. 1.80 m from the

Algorithm: Tracking-by-detection approach

Input: visible Frame = RGB, Thermal Frame = TI, TrackingEnable = False, DetectEnable = True

Output: The final result R

```
for k = 1: N
  if DetectEnable
    roi, Ds = Detection(RGB(k))
    Tracker Initialize(RGB(k), roi)
  end if
  if TrackingEnable
    tracked_roi, Ts = Tracking(RGB(k))
    R(k) = Crop(TI(k), tracked_roi)
    if (Ds && Ts)
      DetectEnable = False
    else
      new_roi, Ds = Detection(RGB(k))
      if Ds
        R(k) = Crop(RGB(k), new_roi)
      else
        R(k) = Crop(TI(k), tracked_roi)
      end if
    end if
  end if
  show(R(k))
end for
```

Fig. 3 – The pseudocode of the tracking-by-detection approach.

subject in uncontrolled room temperature and illumination. Thermal image sequences were acquired using a FLIR E-60 (FLIR Systems Inc., Wilsonville, United States) camera with a resolution of 320 × 240 pixels and a frame rate of 30 Hz. The thermal camera detects the long-wave infrared wavelength (spectral range of 7.5–13 μm) with a thermal sensitivity of less than 0.05 °C at 30 °C. The RGB image sequences were acquired by using a C922 webcam (Logitech International S.A., Lausanne, Switzerland) with a resolution of 960 × 720 pixels and a frame rate of 30 Hz. The visible camera was placed at the top of the thermal camera in a parallel way and this camera setup was mounted on a tripod during video acquisition. A custom-made acquisition algorithm was developed to ensure the simultaneous recording and storing of thermal and visible images. To validate the performance of the proposed method, a Go Direct® Respiration Belt (Vernier Software & Technology, Beaverton, Oregon, United States) was used as the reference measurement. The respiration belt measures the force exerted by the chest or abdomen during respiration with a resolution of 0.01 N [50]. The three phases of the data recording protocol, as described below, were designed for analysis and validation purposes.

Phase 1 (steady): A total of 20 volunteers (6 females) in the age range 28.6 ± 2.6 years participated in Phase 1. In Phase 1, the subjects were sitting still in a chair during recording. However, a slight movement was acceptable. The three different locations were chosen for data recording and the recording time was about 1 min 20 s.

Phase 2 (motion, talking and breathing variation): A total of 12 volunteers (3 females) in the age range 29.2 ± 2.9 years participated in Phase 2. It is further categorised into Phase 2A (motion and talking only) and Phase 2B (motion, talking and breathing variation). During Phase 2A, subjects were free to move their heads in any direction and allowed to talk. Subjects were informed of Phase 2's definition as "data recording with movement permitted" where they cannot stand from a chair and not move out of the FOV. In Phase 2B, they were asked to breathe in various patterns, for example, normal – high breathing – normal – slow breathing – normal. However, there was no fixed time to follow such a pattern. The subjects followed this pattern for a duration and sequence according to their own comfort and understanding. The same three different locations were chosen for data recording. Recording time for Phase 2A was 1 min and 20 s and for Phase 2B was 3 min.

Phase 3 (high temperature in background): A total of 6 volunteers in the age range 29.4 ± 4.2 years participated in Phase 3. In Phase 3, a room heater was put beside the subject to check the efficiency of the proposed method at the high dynamic temperature in the background. A single location was chosen with a recording time of 1 min 20 s.

In general, the physician evaluates the BR by counting the breathing cycle either within 60 s or 30 s. By considering this fact, the BR was evaluated by taking a 30 s window with a step of 20 s. The 20 sec step provides a sufficient gap for successive measurements and also allows the consideration of the subject's facial motion.

In order to evaluate the registration performance, the target registration error (TRE) [51] is employed as follows:

$$TRE = |r_g - r_p| \tag{19}$$

where r_g and r_p denote the ground truth and estimated transformed mapping points, respectively. The nostril detection is quantified by the success rate of ROI detection in each frame [29]. To characterize the error between the proposed non-contact based method and respiration belt, the mean absolute error (MAE) and the standard deviation of error (SDE) have been used. For a measurement 'i', the error metrics are calculated as follows:

$$AE(i) = |BR_m(i) - BR_r(i)| \tag{20}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N} AE(i) \tag{21}$$

$$SDE = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(AE(i) - MAE)^2} \tag{22}$$

where $BR_m(i)$ and $BR_r(i)$ are the breathing rate of the proposed non-contact method and referenced method, respectively. N is the total number of measurements. The root mean square error (RMSE) has also been calculated as per the following expression to evaluate the performance:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(BR_m(i) - BR_r(i))^2} \tag{23}$$

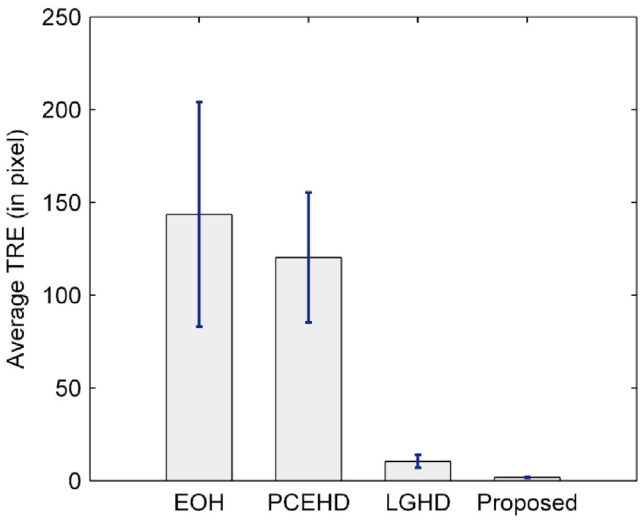


Fig. 4 – Average of target registration errors (TRE) for all 50 RGB-thermal image pairs (The image pair is a randomly selected frame of each recorded video).

The Bland-Altman plot and linear correlation plot have been used to evaluate the effectiveness and the agreement between the proposed non-contact based method and contact based reference method.

4. Results

Prior to the breathing rate analysis, the performance of RGB-thermal image registration and the success of ROI detection was analysed. The proposed registration approach has been compared with numerous standard methods viz. EOH [32], PCEHD [33], LGHD [34], as shown in Fig. 4, and obtained an average TRE of 1.817 pixels with 0.3672 standard error, which is a good performance. The ROI detection rate has been computed for each study phase. The median ROI detection rate was achieved 100% in Phase 1 and Phase 3, 99.87% in Phase 2A and 99.93% in Phase 2B. Videos featuring wearing glasses (n = 8), facial hair (n = 28) and wearing turban (n = 5) scored similar ROI detection rate (99.96%, 99.93% and 100%, respectively).

Fig. 5 shows the boxplot of obtained MAE, SDE, and RMSE between the proposed non-contact BR methods with reference respiratory belt method for each study phase. The value of MAE, SDE and RMSE for each phase are shown in Table A.1 of Appendix A. Fig. 6 depicts the Bland-Altman plot for comparing proposed and reference methods of Phase 1, Phase 2 (Phase 2A and Phase 2B), and Phase 3. An average of RMSE calculated between the proposed and reference method reached 0.1305 ± 0.0741 for Phase 1. Moreover, the MAE was 0.1106 ± 0.0688, and the standard deviation of error (SDE) reached 0.0774 for Phase 1. According to the Bland-Altman plot analysis shown in Fig. 6(a), the mean difference between the proposed and reference methods for Phase 1 was −0.0154 and the limit of agreements ranged from −0.31 bpm to 0.28 bpm. Almost all the data points are within the 95% limit of agreements indicating the proposed method has a good agreement with the reference contact-based method. The observed cor-

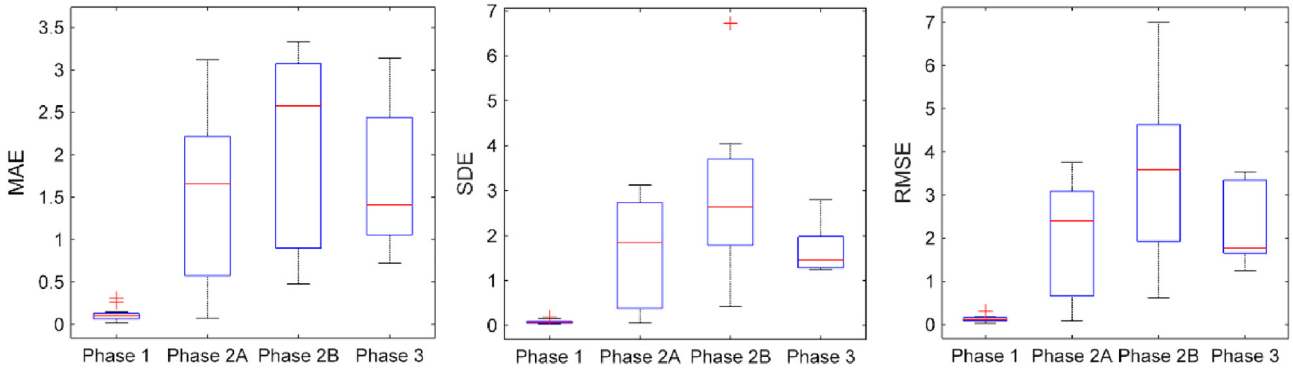


Fig. 5 – The boxplots of obtained MAE, SDE, and RMSE between the proposed non-contact based method and reference respiratory belt method for each phase of the study.

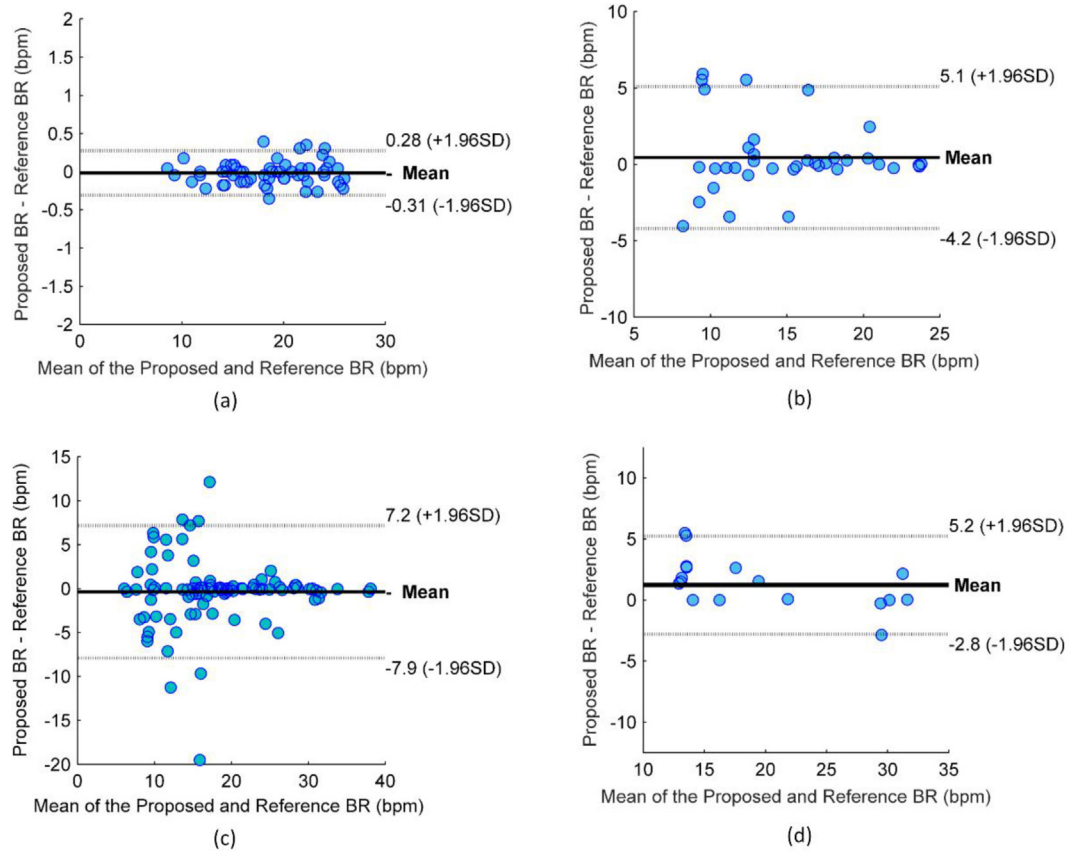


Fig. 6 – Bland-Altman plot of: (a) Phase 1 (N = 60 BR measurements); (b) Phase 2A (N = 36 BR measurements); (c) Phase 2B (N = 96 BR measurements); (d) Phase 3 (N = 18 BR measurements) for comparing proposed non-contact based method and reference respiratory belt method.

relation in the Phase 1 study was 0.9995 with the 95% confidence interval of [0.9991, 0.9997]. To see the effect of different surroundings, the breathing data was recorded in three different locations, namely Place A ('the controlled room with wooden wall'), Place B ('a glass window with curtain'), and Place C ('the angle projection view'). The average MAE and RMSE of three locations for Phase 1 are shown in Fig. 7, demonstrating that the performance is not much affected by different indoor surroundings. Fig. 8 represents the visualization of the breathing wave extraction from automatic detected and tracked ROI in thermal image sequences of the subject. It is contained in Video 1 (see Supplementary Material), aiming to illustrate the working of the proposed method.

The average MAE and SDE between proposed and reference methods were 1.4570 ± 0.9387 and 1.6543 for Phase 2A, respectively, and they were 2.0786 ± 1.0888 and 2.8382 for Phase 2B. Moreover, the average of observed RMSE between the proposed and reference method was 2.0097 ± 1.2580 for Phase 2A and 3.4172 ± 1.7564 for Phase 2B. The corresponding Bland-Altman plots for Phase 2A and Phase 2B subjects are

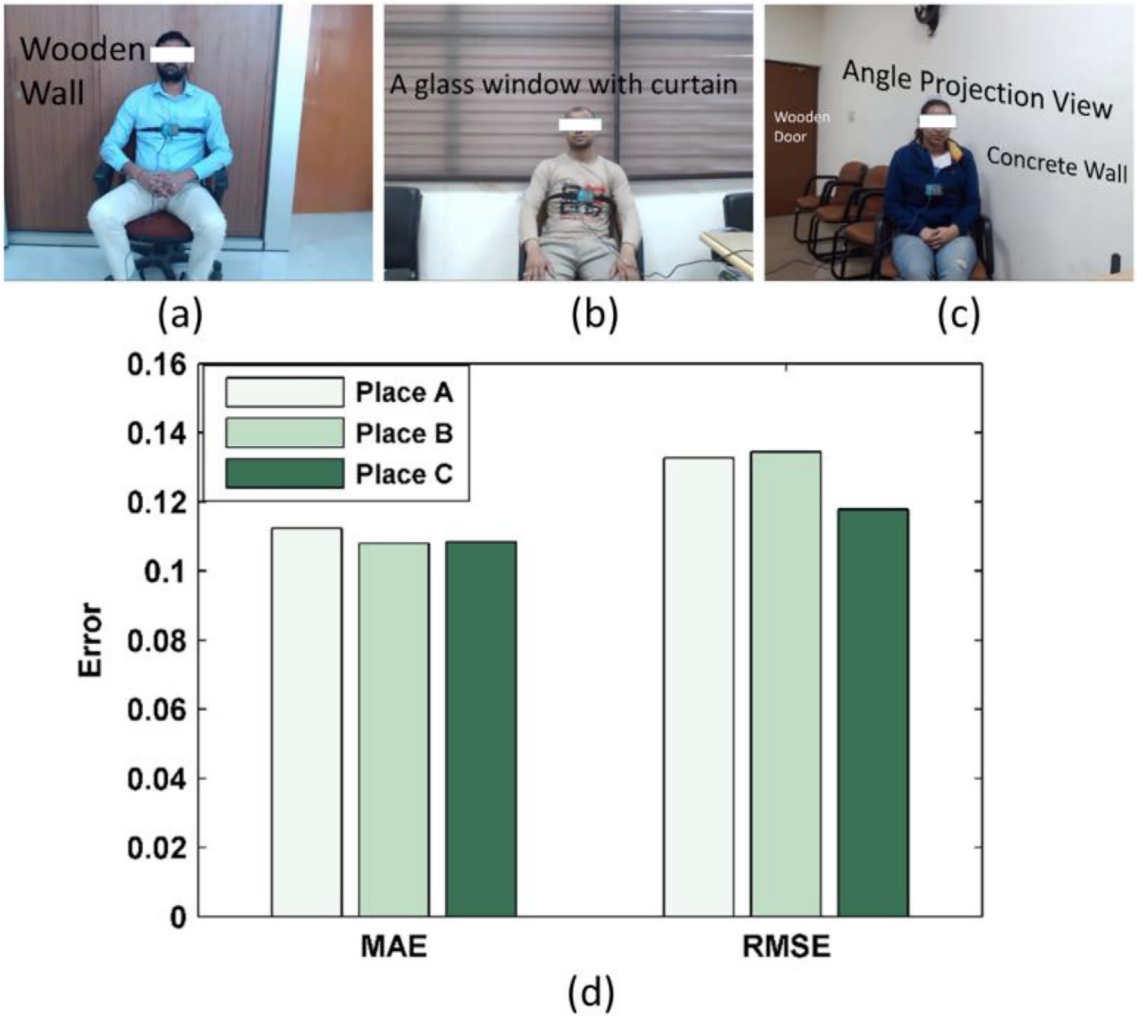


Fig. 7 – Three different locations of data recording: (a) Place A: the controlled room with wooden wall in the background; (b) Place B: a glass window with a curtain in the background, (c) Place C: the angle projection view; (d) The average MAE and RMSE of BR measurements in these locations for Phase 1.

shown in Fig. 6(b) and Fig. 6(c), respectively. Based on the results, the observed mean difference of methods was 0.4568 bpm with the limit of agreement from −4.2 bpm to 5.1 bpm for Phase 2A study, and the mean difference was −0.3589 bpm with the limit of agreements from −7.9 bpm to 7.2 bpm for Phase 2B study. The observed correlation in the Phase 2A study was 0.8766 with the 95% confidence interval of [0.7698; 0.9357], and in Phase 2B was 0.8708 with 95% confidence interval of [0.8121; 0.9120]. Video 2 and Video 3 (see Supplementary Material) illustrate the breathing extraction of the subject in Phase 2A and Phase 2B, respectively. The aim of this visualization is to demonstrate the effectiveness of the proposed work in more practical scenarios where facial movement at a high level, talking, and variable breathing rate are acceptable. Fig. 9 shows the extracted breathing signal and estimated BR from proposed method and reference method of a subject in Phase 2B study.

The estimated average of MAE, SDE, and RMSE were 1.6941 ± 0.9110, 1.7031, and 2.2190 ± 0.9630, respectively, for Phase 3. The Bland-Altman plot in Fig. 6 (d) indicates that the mean difference was 1.231 bpm with the limit of agreement ranging from −2.8 bpm to 5.2 bpm. The observed correlation in the Phase 3 study was 0.9774 with the 95% confidence interval of [0.9390, 0.9917].

Fig. 10 shows the average MAE calculated for male and female participants in each phase of the study. The MAE was found slightly higher in females compared to the male subjects. The suggested reason might be the presence of hair underneath the nose in the male subject. In general, the presence of hair underneath the nose improves the temperature gradient due to respiratory activity. To further analyze it, a comparative plot between subject participated with moustache and without moustache in each phase of the study is shown in Fig. 11. The linear correlation curve and the Bland-Altman plot, which include the BR measurements of all the phases, are shown in Fig. 12. The results show that the proposed non-contact based method has a strong linear correlation with the contact-based reference method. From

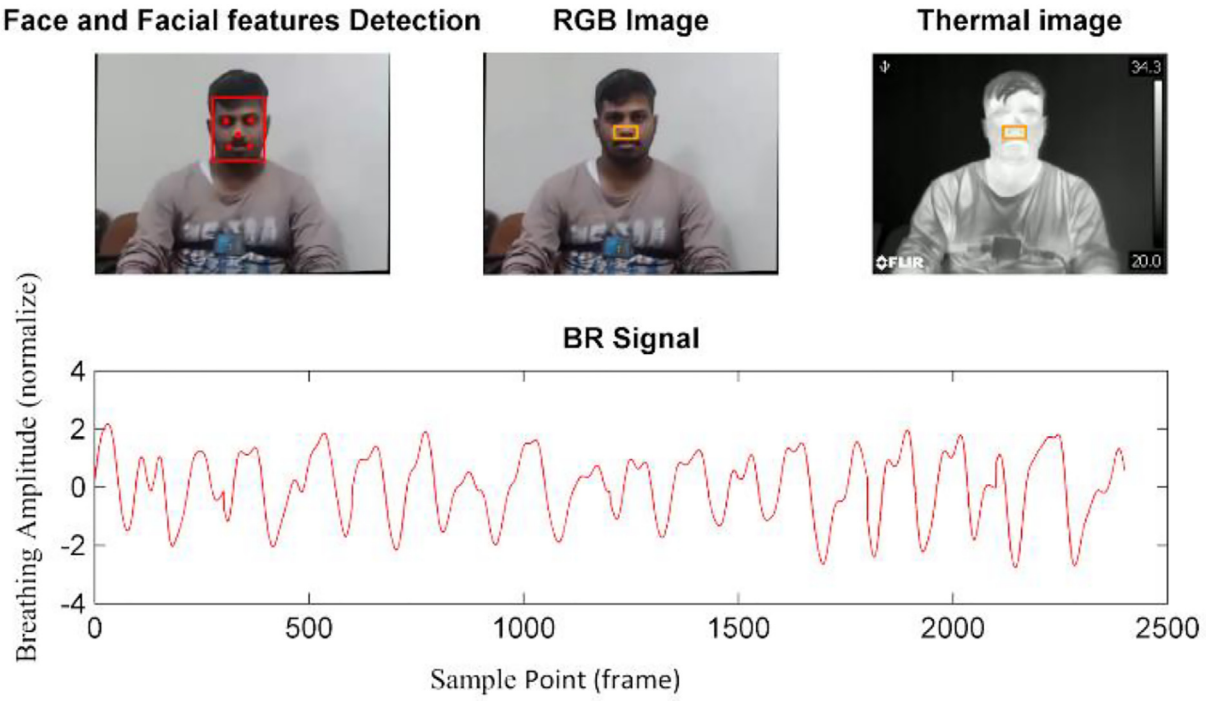


Fig. 8 – The screenshot of breathing signal extraction of the proposed non-contact based method. It is contained in Video 1 that illustrating the effectiveness of the proposed method.

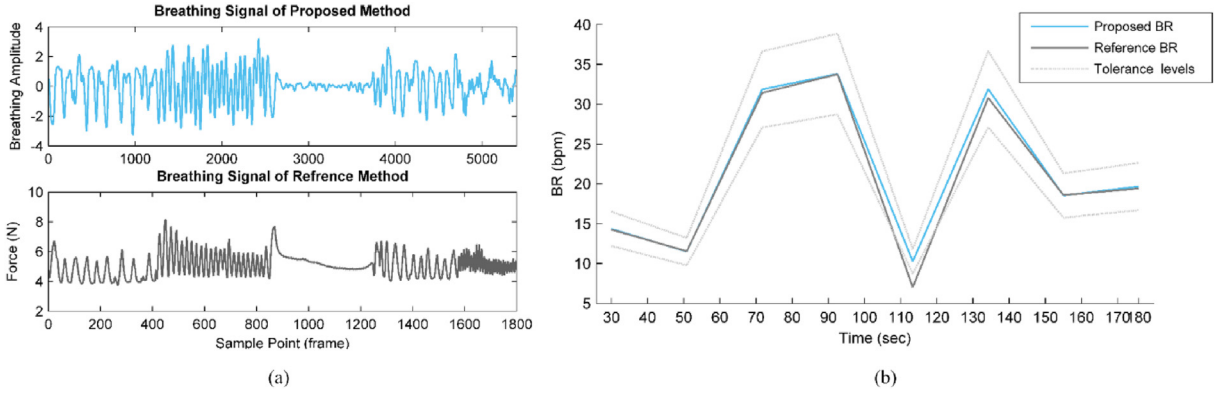


Fig. 9 – (a) Breathing signal (x-axis represents the sample point (frame)); (b) Breathing rate (BR) of a subject from Phase 2B. The tolerance levels are ± 15% (according to the medical practice) of the proposed method's breathing rate.

a combined Bland-Alman plot analysis, the mean difference between the proposed and reference methods for all the sample data was −0.04 and the limit of agreements ranged from −4.1 bpm to 4.0 bpm.

5. Discussion

The results demonstrate that the proposed approach is able to successfully monitor breathing remotely even in most challenging cases, including subject wearing glasses, rapid face movements, talking, variable respiration order, and high thermal dynamic range scene. Also, the proposed method automatically selects the nostril area (ROI) and efficiently performs the tracking of the nostril area, which are the limitations in respiratory monitoring as discussed in [20–23]. The ROI detection rate of the proposed method is excellent, and no significant changes are observed in performance due to wearing glasses, facial hair, and a high temperature range in the background. Only due to the rapid facial movement in Phase 2, slightly lower performance was observed. Table A.1 shows that the mean absolute error is approximately 0.11, and the correlation is 0.9995, which is an excel-

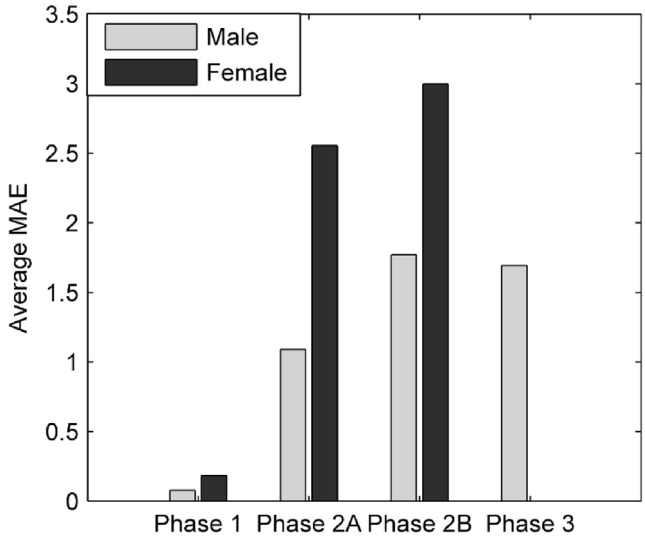


Fig. 10 – The average mean absolute error (MAE) of male and female subjects participated in different phases of the study.

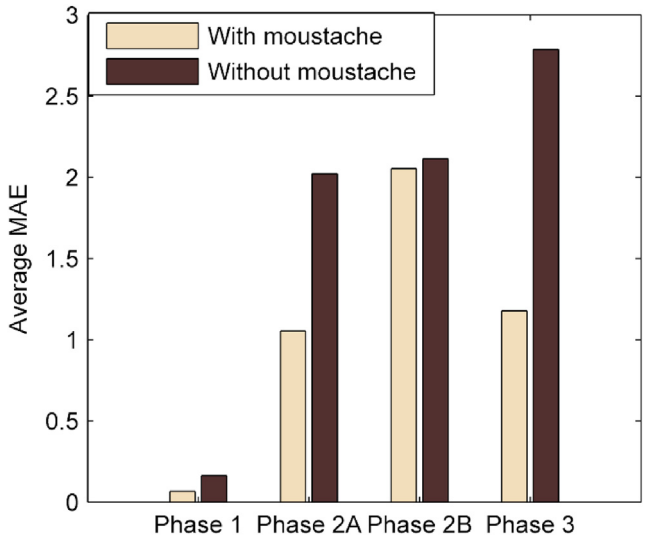


Fig. 11 – The average mean absolute error (MAE) of subjects participated with moustache and without moustache in different phases of the study.

lent agreement in a stable condition of the recording. The motive of the Phase 1 study is to see the effectiveness in standardized clinical scenarios. The findings are similar to the results of Pereira et al. [17], however, they used a high-resolution IR camera (VarioCAM® HD head 820S; resolution 1024 × 768 pixels; sensitivity 0.05 K at 30 °C) and also the selection of ROI was depended on the periorbital region, which is generally a hotspot in the human face. This type of approach might fail in case of high temperatures in the background and wearing glasses.

In Phase 2, the high movement of the face, variant breathing order and talking were allowed to analyse the effectiveness of the proposed method in more practical scenarios such as sport studies, BR during exercise, in home care, and in airport screening. The errors in results are higher as compared to Phase 1, since the motion and disturbance of air from nostrils interfere with the breathing pattern. Fig. 9 demonstrates that, like the reference method, the derived breathing signal and BR from the proposed method follow the variation in breathing patterns. The proposed method has an effective agreement with the reference method and has less error during motion and talking than the study of a combination of breathing during paddling and talking scenarios by Chauvin et al. [52]. In contrast to our method, Chauvin et al. used a high-resolution IR camera (IRXCAM-640; resolution 640 × 480 pixels) and the initial selection of ROI was manual using a graphical user interface (GUI). Compared to the previously published work [17,52], the use of visible image sequence for the ROI selection and tracking, and the linear mapping with thermal images by using image registration compensate for the limitation of manual ROI selection and poor tracking in the practical scenario of motion and talking.

In the proposed method, automatic image registration is used, which is independent of camera parameters and provides flexibility in camera setup. The methods proposed by Hu et al. [31] and Sceeba et al. [29] uses a fixed imaging setup and the registration parameters are obtained either by selecting strong correspondence points from the first frame of both RGB-thermal video pairs [31] or during the calibration phase and applied the same parameters to all the recorded video for linear mapping [29,30]. This type of fixed manual registration is greatly influenced by the camera-to-subject distance and results in poor linear mapping. By contrast, in the proposed method, first, the scale adjustment pre-transformation is applied when the camera is setup at a particular place. After that, the fine registration approach automatically selects the correspondence pairs according to image pairs. The fine registration tunes the registration parameters for effective liner mapping between RGB and thermal image for each recording.

One of the major limitations in the case of thermal imaging based method is the obstacle of nose (ROI) and moving out of the FOV during recording, as discussed in [31]. In the existing methods [29,30], the frame was discarded where the detector failed to detect ROI and the detector was reinitialized every fixed time duration. The tracking-by-detection method in the proposed approach automatically reinitializes the ROI detection and performs tracking successively. Therefore, when the obstacle becomes clear, the breathing pattern is successfully recorded again. In the Phase 2 study, despite strong head rotation (up-, down- left- and rightwards) and the obstacle between ROI, the detection-of-tracking method has efficiently followed the ROI.

The infrared camera used in this work performs non-uniformity correction (NUC) automatically. The camera software uses NUC to compensate for the sudden temperature deviations and geometrical disturbances [53]. During the NUC, image acquisition freezes for a short time and interrupts the breathing signal. However, this has no effect on ROI selection and tracking, since the RGB images are used for the ROI selection and tracking. By selecting the large window of 30 sec in BR measurement, the small interruption in temperature recording caused by the NUC might be balanced out.

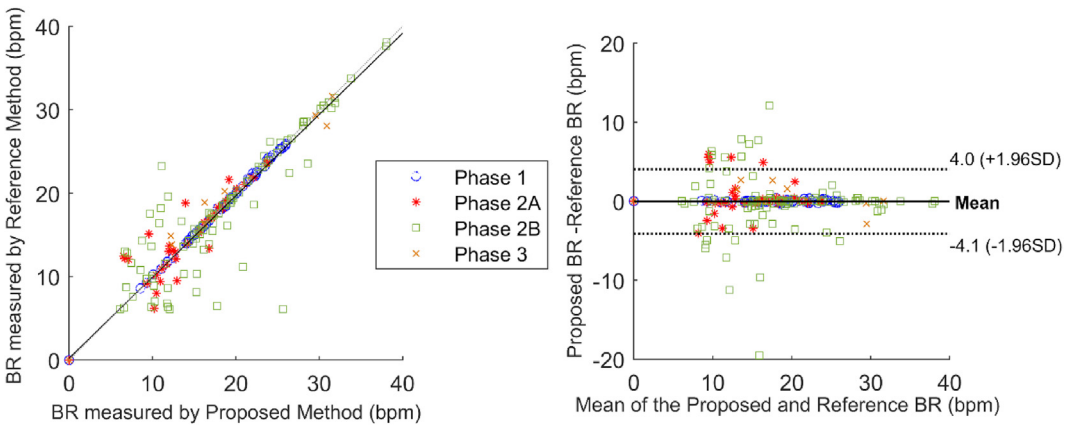


Fig. 12 – The combined liner correlation curve (left) and Bland-Altman plot (right) for all the phases of study. (Total N = 210 BR measurements). The different colors and shapes are used to represent the BR measurements of different phases.

6. Conclusion

In recent decades, the demand for non-contact breathing rate monitoring has grown rapidly as it is safe, comfortable and improves the quality of life. Also, it is useful in the COVID-19 pandemic situation. This paper introduces a novel, robust, and feasible non-contact BR detection method using RGB-thermal imaging. The RGB and thermal image sequences are acquired simultaneously and integrated with the proposed RGB-thermal image registration. The proposed methodology combined RGB-thermal image registration, tracking-by-detection, and breathing rate extraction to estimate the breathing rate accurately and achieve a good agreement with the ground truth. Moreover, the experimental results show that the proposed method is successfully estimated in more challenging situations such as fast motion, talking, and high background temperature. In the future, the effectiveness and feasibility of real-time and long-time monitoring will be investigated. Simultaneous breathing rate extraction of multiple people with different assumptions would be considered in the future to increase the proposed method's importance.

7. Informed consent Statement

Informed consent was obtained from all the subjects involved in the study. Also, consent was gained from the subjects for the publication of their videos and facial images.

Ethical Statement

Before participating in the study, all subjects gave their informed consent to be included. The study was carried out in accordance with the Declaration of Helsinki, and the methodology and protocol were approved in the Institution (CSIR-Central Scientific Instruments Organisation, Chandigarh, India, https://www.csio.res.in/) having the Ethical Committee. Approval number Ref. IEC/CSIO/2020/Project No. 06; Dated 9th January 2020.

Author contribution

L. Maurya conceptualised and designed the experiment, collected data and conducted the experiment, analysed the data, and drafted the manuscript. P. Mahapatra oversaw the work, provided resources and laboratory facilities, and analysed the manuscript. D. Chawla contributed to clinical aspects of technical challenges and opportunities. P. Mahapatra and D. Chawla supervised the entire work and critically reviewed the final draft. All the authors have read and approved the manuscript for submission.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

See Table A.1

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbe.2021.07.002.

Table A.1 – The MAE, SDE, and RMSE value for each phase of study.

S.N.	Phase 1				Phase 2							Phase 3			
	Subject				Subject	Phase 2A			Phase 2B			Subject			
		MAE	SDE	RMSE		MAE	SDE	RMSE	MAE	SDE	RMSE		MAE	SDE	RMSE
1	P1_1	0.0586	0.1015	0.1015	P2_1	0.2053	0.0672	0.2125	0.6598	1.1084	1.2289	P3_1	1.3930	1.3446	1.7736
2	P1_2	0.1318	0.1318	0.1702	P2_2	0.3267	0.0966	0.3361	1.2360	2.6741	2.7901	P3_2	1.0547	1.5642	1.6564
3	P1_3	0.0879	0.0761	0.1076	P2_3	0.9814	1.2815	1.4346	0.7093	1.1912	1.3208	P3_3	1.4223	1.2797	1.7648
4	P1_4	0.1318	0.1163	0.1625	P2_4	1.6569	2.0793	2.3722	2.5708	2.6126	3.5471	P3_4	2.4389	2.7998	3.3427
5	P1_5	0.0732	0.0254	0.0761	P2_5	2.0508	1.6214	2.4410	2.8537	2.3784	3.6185	P3_5	0.7212	1.2491	1.2491
6	P1_6	0.1172	0.0507	0.1243	P2_6	3.1201	2.5846	3.7668	3.0817	4.0471	4.8814	P3_6	3.1348	1.9811	3.5275
7	P1_7	0.1172	0.0915	0.1390	P2_7	1.6553	2.7912	2.8167	3.2849	4.0162	4.9904				
8	P1_8	0.1172	0.0507	0.1243	P2_8	2.4927	2.7224	3.3398	3.0652	6.7225	6.9956				
9	P1_9	0.0879	0.0761	0.1076	P2_9	0.8203	0.6669	0.9846	2.5818	3.4022	4.0980				
10	P1_10	0.0146	0.0254	0.0254	P2_10	1.7285	2.7659	2.8439	3.3289	3.0266	4.3700				
11	P1_11	0.0587	0.0254	0.0622	P2_11	0.0733	0.0508	0.0842	1.0931	2.4587	2.5465				
12	P1_12	0.1318	0.0879	0.1501	P2_12	2.3730	3.1238	3.4838	0.4779	0.4205	0.6190				
13	P1_13	0.1465	0.0671	0.1564											
14	P1_14	0.1318	0.1522	0.1812											
15	P1_15	0.0587	0.0672	0.0803											
16	P1_16	0.3076	0.0439	0.3097											
17	P1_17	0.0733	0.0508	0.0842											
18	P1_18	0.0293	0.0254	0.0359											
19	P1_19	0.0732	0.0915	0.1046											
20	P1_20	0.2637	0.1916	0.3066											
Mean	**Total = 20**	**0.1106**	**0.0774**	**0.1305**	**Total = 12**	**1.4570**	**1.6543**	**2.0097**	**2.0786**	**2.8382**	**3.4172**	**Total = 6**	**1.6941**	**1.7031**	**2.2190**

REFERENCES

[1] I. Wheatley, Respiratory rate 3: how to take an accurate measurement n.d. https://www.nursingtimes.net/clinical-archive/respiratory-clinical-archive/respiratory-rate-3-how-to-take-an-accurate-measurement-25-06-2018/ (accessed April 14, 2021).

[2] Edwards MO, Kotecha SJ, Kotecha S. Respiratory Distress of the Term Newborn Infant. Paediatr. Respir. Rev. 2013;14 (1):29–37. https://doi.org/10.1016/j.prrv.2012.02.002.

[3] A.D. Droitcour, T.B. Seto, B.K. Park, S. Yamada, A. Vergara, C. El Hourani, et al. Non-contact respiratory rate measurement validation for hospitalized patients, in: Proc. 31st Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. Eng. Futur. Biomed. EMBC 2009, vol. 2009, IEEE Computer Society; 2009, p. 4812–5. https://doi.org/10.1109/IEMBS.2009.5332635.

[4] Cretikos MA, Bellomo R, Hillman K, Chen J, Finfer S, Flabouris A. Respiratory rate: The neglected vital sign. Med. J. Aust. 2008;188:657–9. https://doi.org/10.5694/j.1326-5377.2008.tb01825.x.

[5] Prince PGK, Immanuel Rajkumar R, Premalatha J. Novel Non-Contact Respiration Rate Detector for Analysis of Emotions. Cham: Springer; 2020. p. 157–78. https://doi.org/10.1007/978-3-030-35139-7_8.

[6] Darzi A, Goršič M, Novak D. Difficulty adaptation in a competitive arm rehabilitation game using real-time control of arm electromyogram and respiration. IEEE Int. Conf. Rehabil. Robot., IEEE Comput. Soc. 2017:857–62. https://doi.org/10.1109/ICORR.2017.8009356.

[7] Sun G, Nakayama Y, Dagdanpurev S, Abe S, Nishimura H, Kirimoto T, et al. Remote sensing of multiple vital signs using a CMOS camera-equipped infrared thermography system and its clinical application in rapidly screening patients with suspected infectious diseases. Int. J. Infect. Dis. 2017;55:113–7. https://doi.org/10.1016/j.ijid.2017.01.007.

[8] Jiang Z, Hu M, Gao Z, Fan L, Dai R, Pan Y, et al. Detection of respiratory infections using RGB-infrared sensors on portable device. IEEE Sens. J. 2020;20:13674–81. https://doi.org/10.1109/JSEN.2020.3004568.

[9] Naranjo J, Centeno RA, Galiano D, Beaus M. A nomogram for assessment of breathing patterns during treadmill exercise. Br. J. Sports Med. 2005;39:80–3. https://doi.org/10.1136/bjsm.2003.009316.

[10] Flenady T, Dwyer T, Applegarth J. Accurate respiratory rates count: So should you! Australas Emerg. Nurs. J. 2017;20:45–7. https://doi.org/10.1016/j.aenj.2016.12.003.

[11] Baharestani MM. An overview of neonatal and pediatric wound care knowledge and considerations. Ostomy Wound Manag. 2007;53:34–55.

[12] Droitcour AD, Boric-Lubecke O, Kovacs GTA. Signal-to-noise ratio in doppler radar system for heart and respiratory rate measurements. IEEE Trans. Microw. Theory Tech. 2009;57:2498–507. https://doi.org/10.1109/TMTT.2009.2029668.

[13] Procházka A, Schätz M, Vyšata O, Vališ M. Microsoft kinect visual and depth sensors for breathing and heart rate analysis. Sensors 2016;16:996. https://doi.org/10.3390/s16070996.

[14] Wang Y, Hu M, Zhou Y, Li Q, Yao N, Zhai G, et al. Unobtrusive and automatic classification of multiple people's abnormal respiratory patterns in real time using deep neural network and depth camera. IEEE Internet Things J. 2020;7:8559–71. https://doi.org/10.1109/JIOT.2020.2991456.

[15] M. Bartula, T. Tigges, J. Muehlsteff, Camera-based system for contactless monitoring of respiration, in: Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. EMBS, vol. 2013, Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.; 2013, p. 2672–5. https://doi.org/10.1109/EMBC.2013.6610090.

[16] van Gastel M, Stuijk S, de Haan G. Robust respiration detection from remote photoplethysmography. Biomed. Opt. Express 2016;7:4941. https://doi.org/10.1364/BOE.7.004941.

[17] Pereira CB, Yu X, Czaplik M, Rossaint R, Blazek V, Leonhardt S. Remote monitoring of breathing dynamics using infrared thermography. Biomed. Opt. Express 2015;6:4378. https://doi.org/10.1364/BOE.6.004378.

[18] Al-Naji A, Gibson K, Lee SH, Chahl J. Monitoring of cardiorespiratory signal: principles of remote measurements and review of methods. IEEE Access 2017;5:15776–90. https://doi.org/10.1109/ACCESS.2017.2735419.

[19] Massaroni C, Nicolo A, Sacchetti M, Schena E. Contactless methods for measuring respiratory rate: a review. IEEE Sens. J. 2021;21:12821–39. https://doi.org/10.1109/JSEN.2020.3023486.

[20] Z. Zhu, J. Fei, I. Pavlidis, Tracking human breath in infrared imaging, in: Proc. - BIBE 2005 5th IEEE Symp. Bioinforma. Bioeng., vol. 2005, 2005, p. 227–31. https://doi.org/10.1109/BIBE.2005.55.

[21] Murthy R, Pavlidis I. Noncontact measurement of breathing function. IEEE Eng. Med. Biol. Mag. 2006;25:57–67. https://doi.org/10.1109/MEMB.2006.1636352.

[22] Abbas AK, Heiman K, Orlikowsky T, Leonhardt S. Non-contact respiratory monitoring based on real-time IR-thermography. In: IFMBE Proc.. Springer Verlag; 2009. p. 1306–9. https://doi.org/10.1007/978-3-642-03882-2_346.

[23] AL-Khalidi F, Saatchi R, Elphick H, Burke D, Hisham. An evaluation of thermal imaging based respiration rate monitoring in children. Am. J. Eng. Appl. Sci. 2011;4:586–97. https://doi.org/10.3844/ajeassp.2011.586.597.

[24] Barbosa Pereira C, Yu X, Czaplik M, Blazek V, Venema B, Leonhardt S. Estimation of breathing rate in thermal imaging videos: a pilot study on healthy human subjects. J. Clin. Monit. Comput. 2017;31:1241–54. https://doi.org/10.1007/s10877-016-9949-y.

[25] Cho Y, Julier SJ, Marquardt N, Bianchi-Berthouze N. Robust tracking of respiratory rate in high- dynamic range scenes using mobile thermal imaging. ArXiv 2017;20:90782. https://doi.org/10.1364/BOE.8.004480.

[26] Jagadev P, Giri LI. Non-contact monitoring of human respiration using infrared thermography and machine learning. Infrared Phys. Technol. 2020;104:103117. https://doi.org/10.1016/j.infrared.2019.103117.

[27] Ma J, Zhou Z, Wang B, Zong H. Infrared and visible image fusion based on visual saliency map and weighted least square optimization. Infrared Phys. Technol. 2017;82:8–17. https://doi.org/10.1016/j.infrared.2017.02.005.

[28] Wang J-G, Sung E. Facial feature extraction in an infrared image by proxy with a visible face image. IEEE Trans. Instrum. Meas. 2007;56:2057–66. https://doi.org/10.1109/TIM.2007.904567.

[29] G. Scebba, L. Tüshaus, W. Karlen, Multispectral camera fusion increases robustness of ROI detection for biosignal estimation with nearables in real-world scenarios, in: 2018 40th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc., 2018, p. 5672–5. https://doi.org/10.1109/EMBC.2018.8513501.

[30] Scebba G, Da Poian G, Karlen W. Multispectral video fusion for non-contact monitoring of respiratory rate and apnea. IEEE Trans. Biomed. Eng. 2021;68:350–9. https://doi.org/10.1109/TBME.1010.1109/TBME.2020.2993649.

[31] Hu M-H, Zhai G-T, Li D, Fan Y-Z, Chen X-H, Yang X-K. Synergetic use of thermal and visible imaging techniques for contactless and unobtrusive breathing measurement. J. Biomed. Opt. 2017;22:1. https://doi.org/10.1117/1.JBO.22.3.036006.

[32] Aguilera C, Barrera F, Lumbreras F, Sappa AD, Toledo R. Multispectral image feature points. Sensors 2012;12:12661–72. https://doi.org/10.3390/s120912661.
[33] Mouats T, Aouf N, Sappa AD, Aguilera C, Toledo R. Multispectral stereo odometry. IEEE Trans. Intell. Transp. Syst. 2015;16:1210–24. https://doi.org/10.1109/TITS.2014.2354731.
[34] C.A. Aguilera, A.D. Sappa, R. Toledo, LGHD: A feature descriptor for matching across non-linear intensity variations, in: 2015 IEEE Int. Conf. Image Process., 2015, p. 178–81. https://doi.org/10.1109/ICIP.2015.7350783.
[35] Ye Y, Shen L, Hao M, Wang J, Xu Z. Robust optical-to-SAR image matching based on shape properties. IEEE Geosci. Remote Sens. Lett. 2017;14:564–8. https://doi.org/10.1109/LGRS.2017.2660067.
[36] Li J, Hu Q, Ai M. RIFT: multi-modal image matching based on radiation-variation insensitive feature transform. IEEE Trans. Image Process. 2020;29:3296–310. https://doi.org/10.1109/TIP.8310.1109/TIP.2019.2959244.
[37] Hu M, Zhai G, Li D, Fan Y, Duan H, Zhu W, et al. Combination of near-infrared and thermal imaging techniques for the remote and simultaneous measurements of breathing and heart rates under sleep situation. PLoS ONE 2018;13. https://doi.org/10.1371/journal.pone.0190466 e0190466.
[38] L. Maurya, P. Mahapatra, D. Chawla, S. Verma, An automatic thermal and visible image registration using a calibration rig. Adv. Intell. Syst. Comput., vol. 1124, Springer; 2020, p. 67–76. https://doi.org/10.1007/978-981-15-2740-1_5.
[39] Morrone MC, Owens RA. Feature detection from local energy. Pattern Recogn. Lett. 1987;6:303–13. https://doi.org/10.1016/0167-8655(87)90013-4.
[40] Kovesi P. Phase congruency: A low-level image invariant. Psychol. Res. 2000;64:136–48. https://doi.org/10.1007/s004260000024.
[41] Horn BK. Robot Vision. 1st ed. McGraw-Hill Higher Education; 1986.
[42] Rosten E, Porter R, Drummond T. Faster and better: A machine learning approach to corner detection. IEEE Trans. Pattern Anal. Mach. Intell. 2010;32:105–19. https://doi.org/10.1109/TPAMI.2008.275.
[43] Wu Y, Ma W, Gong M, Su L, Jiao L. A novel point-matching algorithm based on fast sample consensus for image registration. IEEE Geosci. Remote Sens. Lett. 2015;12:43–7. https://doi.org/10.1109/LGRS.2014.2325970.
[44] Zhang K, Zhang Z, Li Z, Qiao Y. Joint face detection and alignment using multitask cascaded convolutional networks. IEEE Signal Process Lett. 2016;23:1499–503. https://doi.org/10.1109/LSP.2016.2603342.
[45] Liu S, Liu D, Srivastava G, Połap D, Woźniak M. Overview and methods of correlation filter algorithms in object tracking. Complex Intell Syst 2020;1:3. https://doi.org/10.1007/s40747-020-00161-4.
[46] M. Danelljan, G. Häger, F.S. Khan, M. Felsberg, Accurate scale estimation for robust visual tracking, in: BMVC 2014 - Proc. Br. Mach. Vis. Conf. 2014, British Machine Vision Association, BMVA; 2014. https://doi.org/10.5244/c.28.65.
[47] Leys C, Ley C, Klein O, Bernard P, Licata L. Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median. J. Exp. Soc. Psychol. 2013;49:764–6. https://doi.org/10.1016/j.jesp.2013.03.013.
[48] Rabiner L, Schafer R, Rader C. The chirp z-transform algorithm. IEEE Trans. Audio Electroacoust. 1969;17:86–92. https://doi.org/10.1109/TAU.1969.1162034.
[49] Hsieh C-H, Chiu Y-F, Shen Y-H, Chu T-S, Huang Y-H. A UWB radar signal processing platform for real-time human respiratory feature extraction based on four-segment linear waveform model. IEEE Trans. Biomed. Circuits Syst. 2016;10:219–30. https://doi.org/10.1109/TBCAS.2014.2376956.
[50] Go Direct® Respiration Belt User Manual – Vernier n.d. https://www.vernier.com/manuals/gdx-rb/ (accessed April 14, 2021).
[51] Maurer CR, Maciunas RJ, Fitzpatrick JM. Registration of head CT images to physical space using a weighted combination of points and surfaces [image-guided surgery]. IEEE Trans. Med. Imaging 1998;17:753–61. https://doi.org/10.1109/42.736031.
[52] Chauvin R, Hamel M, Briere S, Ferland F, Grondin F, Letourneau D, et al. Contact-free respiration rate monitoring using a pan-tilt thermal camera for stationary bike telerehabilitation sessions. IEEE Syst. J. 2016;10:1046–55. https://doi.org/10.1109/JSYST.2014.2336372.
[53] User's manual FLIR Exx series n.d. https://www.flir.com/globalassets/imported-assets/document/flir-exx-series-user-manual.pdf (accessed April 14, 2021).